

SAMPLE - NOT ISSUED. No Engagement Terms are incorporated and no commitment of any kind is made. Issued reports carry the governing terms version here.

SIRPY Verified — Tier 1 Report

Raw computational output with the full honesty floor. This tier makes no formal certification claims; it reports what was measured.

Order identity

Field	Value
Client	Acme Labs
Order id	LQI-2026-Troesch-20-2026-08-29-T1
Reference	LQI-2026-Troesch-20
Problem	A plasma-confinement model.
Stated equation & data	$y'' = 20 \cdot \sinh(20 \cdot y)$, $y(0) = 0$, $y(1) = 1$
Ordered (date in)	2026-08-29
Delivered (date out)	2026-08-29
Service tier	Tier 1: Verified

Run identity and finding

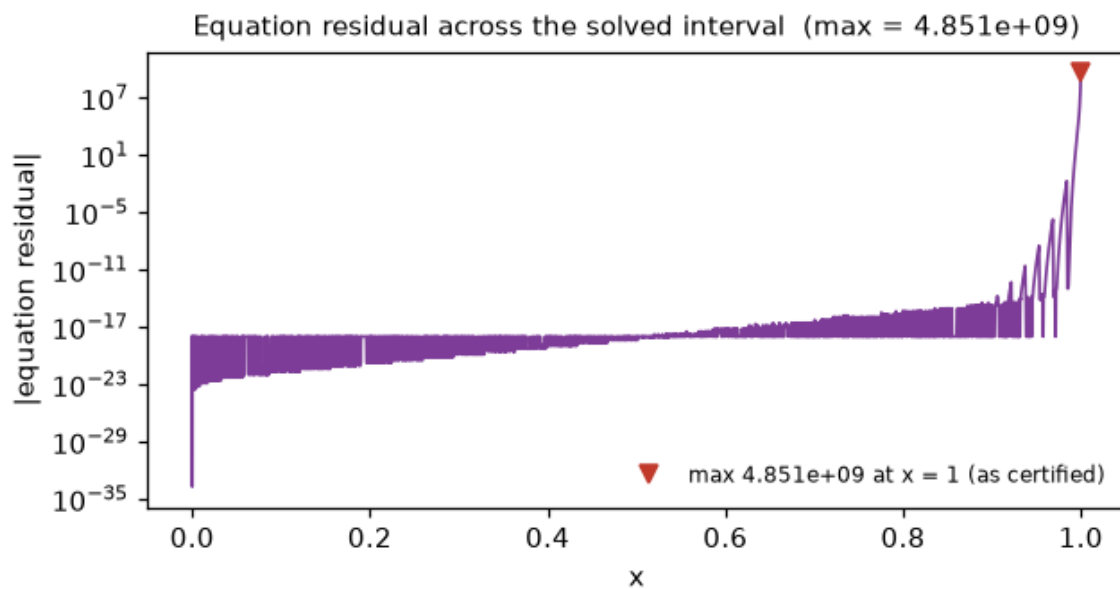
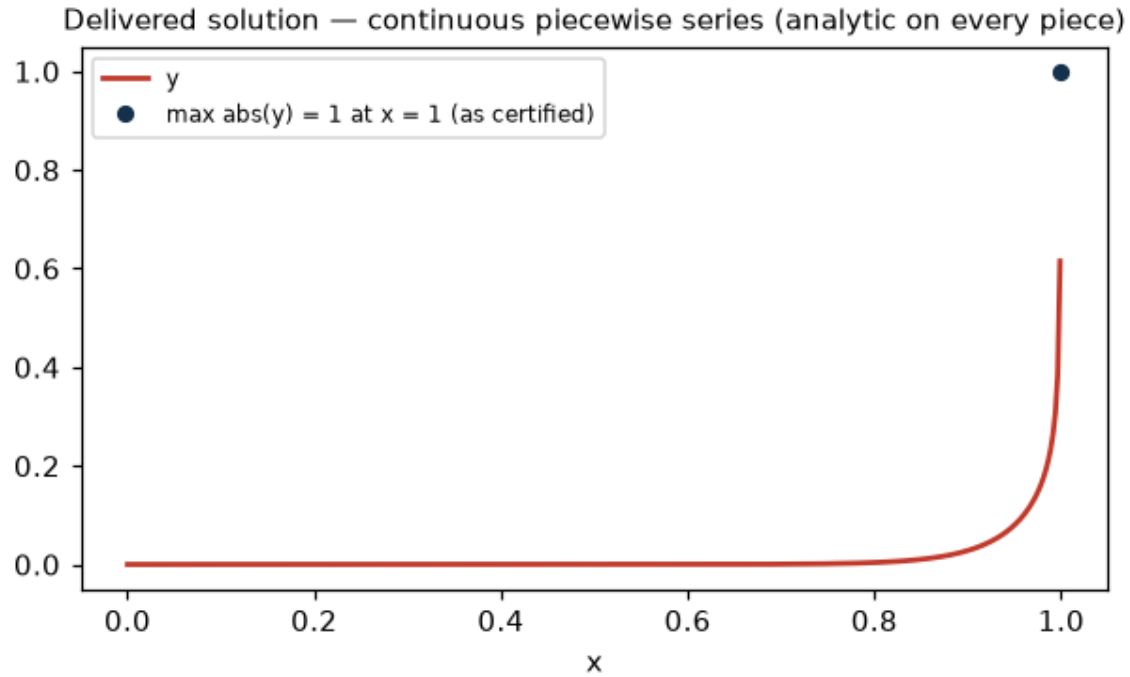
- Class: ODE boundary-value problem | Engine: SIRPY native — power series | Generated by SIRPY 0.59.15
- FINDING: solved, NOT verified
- max residual (measured): 4.851e+09 at $x = 1$
- artifact coverage: [0, 1]; pieces solved 64, delivered 64
- artifact: LQI-2026-Troesch-20-T1.artifact.json
- sha256: 3e50acb4 28d92f15 1d01d3d7 f5a1c638 87d32551 364f55b3 bd4c4d2a a772e726
- reproduction: the headline numbers above are RECOMPUTABLE from this artifact and the stated equation using only Python and sha256sum -- substitute the delivered coefficients into the equation and compare. This is reproduction of our measurements, not agreement with a third-party solver: an independent solver may be compared against the delivered values, and any disagreement is a finding to report, not a covered reproducibility failure. Scope: the artifact coverage above.

Verify facts (measured on this run)

Item	Value
checked	the equation (residual by exact coefficient differentiation)
checked	boundary data
max equation residual	4.85125e+09

Item	Value
at	1
boundary/initial residual	7.74936e-14

Figures



Result

SEE FINDING ABOVE

Artifact: LQI-2026-Troesch-20-T1.artifact.json — sha256

3e50acb428d92f151d01d3d7f5a1c63887d32551364f55b3bd4c4d2aa772e726. Every number above is reproducible from that file and the stated equation using Python and sha256sum alone.

Issued by LQI. The order manifest (ORDER_AND_DELIVERY.txt) is included in this delivery package alongside this document and the artifact; the terms of service applicable to this order are those supplied with your order confirmation and named by version in the delivery manifest.